

Data Requirements Identification -Task-2

AJIO Products Order Booking Database System

Student Name: Harish SP

Register Number: ASML25005

1. Entity List with Attributes and Primary Keys

Entity 1. Customer

Attribute Name	Data Type	Description	PK
CustomerID	INT	Unique identifier	PK
FirstName	VARCHAR(50)	First name	
LastName	VARCHAR(50)	Last name	
Email	VARCHAR(100)	Unique login email	
Password	VARCHAR(255)	Encrypted password	
Phone	VARCHAR(15)	Unique mobile number	
DateOfBirth	DATE	Date of birth	
RegistrationDate	DATE TIME	Account creation date	
AccountStatus	ENUM('Active','Inactive')	Current status	
LoyaltyPoints	INT	Reward points earned	

Entity 2. Address

Attribute Name	Data Type	Description	PK
AddressID	INT	Unique identifier	PK
CustomerID	INT	References Customer	FK
AddressType	ENUM('Home','Office')	Type of address	
StreetAddress	VARCHAR(255)	House/street details	
City	VARCHAR(50)	City name	
State	VARCHAR(50)	State name	
Pincode	VARCHAR(10)	Postal code	
IsDefault	BOOLEAN	Default shipping address	

Entity 3. Category

Attribute Name	Data Type	Description	PK
CategoryID	INT	Unique identifier	PK
CategoryName	VARCHAR(100)	Category name	
Description	TEXT	Category details	
ParentCategoryID	INT	References parent category	FK
IsActive	BOOLEAN	Active status	

Entity 4. Product

Attribute Name	Data Type	Description	PK
ProductID	INT	Unique identifier	PK
ProductName	VARCHAR(200)	Product title	
Description	TEXT	Detailed description	
CategoryID	INT	References Category	FK
BrandName	VARCHAR(100)	Brand name	
Color	VARCHAR(50)	Product color	
Size	VARCHAR(20)	Size (S, M, L, XL)	

Attribute Name	Data Type	Description	PK
Price	DECIMAL(10,2)	Selling price	
StockQuantity	INT	Available inventory	
ReorderLevel	INT	Minimum stock alert	
IsActive	BOOLEAN	Available for sale	

Entity 5. ProductImage

Attribute Name	Data Type	Description	PK
ImageID	INT	Unique identifier	PK
ProductID	INT	References Product	FK
ImageURL	VARCHAR(255)	Image path	
IsPrimary	BOOLEAN	Main image flag	

Entity 6. Cart

Attribute Name	Data Type	Description	PK
CartID	INT	Unique identifier	PK
CustomerID	INT	References Customer	FK
ProductID	INT	References Product	FK
Quantity	INT	Number of units	
AddedDate	DATETIME	Date added to cart	

Entity 7. Order

Attribute Name	Data Type	Description	PK
OrderID	INT	Unique identifier	PK
CustomerID	INT	References Customer	FK
OrderDate	DATETIME	Order placement date	
OrderStatus	ENUM('Pending','Confirmed','Shipped','Delivered','Cancelled')	Current status	
TotalAmount	DECIMAL(10,2)	Total order value	
PaymentMethod	ENUM('Card','UPI','COD','Wallet')	Payment type	
ShippingAddressID	INT	References Address	FK

Entity 8. OrderItem

Attribute Name	Data Type	Description	PK
OrderItemID	INT	Unique identifier	PK
OrderID	INT	References Order	FK
ProductID	INT	References Product	FK
Quantity	INT	Units purchased	
UnitPrice	DECIMAL(10,2)	Price at purchase time	
TotalPrice	DECIMAL(10,2)	Total line amount	

Entity 9. Payment

Attribute Name	Data Type	Description	PK
PaymentID	INT	Unique identifier	PK
OrderID	INT	References Order	FK
Amount	DECIMAL(10,2)	Payment amount	
PaymentDate	DATETIME	Payment date/time	
PaymentStatus	ENUM('Pending','Successful','Failed','Refunded')	Current status	
TransactionID	VARCHAR(100)	Gateway transaction ID	

Entity 10. Shipment

Attribute Name	Data Type	Description	PK
ShipmentID	INT	Unique identifier	PK
OrderID	INT	References Order	FK
CourierName	VARCHAR(100)	Courier service	
TrackingNumber	VARCHAR(100)	Tracking ID	
ShipmentDate	DATETIME	Dispatch date	
DeliveryStatus	ENUM('Pending','In Transit','Delivered','Returned')	Delivery status	

Entity 11. Review

Attribute Name	Data Type	Description	PK
ReviewID	INT	Unique identifier	PK
CustomerID	INT	References Customer	FK
ProductID	INT	References Product	FK
Rating	INT	1-5 rating	
ReviewContent	TEXT	Review text	
ReviewDate	DATETIME	Submission date	

Entity 12. Wishlist

Attribute Name	Data Type	Description	PK
WishlistID	INT	Unique identifier	PK
CustomerID	INT	References Customer	FK
ProductID	INT	References Product	FK
AddedDate	DATETIME	Date added	

Entity 13. AuditLog

Attribute Name	Data Type	Description	PK
LogID	INT	Unique identifier	PK
UserID	INT	User who performed action	
ActionType	VARCHAR(50)	Type of action	
TableName	VARCHAR(50)	Table affected	
OldValue	TEXT	Previous value	
NewValue	TEXT	New value	
ActionDate	DATETIME	Action timestamp	

2. Primary Key Summary

Entity Name	Primary Key	Data Type
Customer	CustomerID	INT
Address	AddressID	INT
Category	CategoryID	INT
Product	ProductID	INT
ProductImage	ImageID	INT
Cart	CartID	INT
Order	OrderID	INT
OrderItem	OrderItemID	INT
Payment	PaymentID	INT
Shipment	ShipmentID	INT
Review	ReviewID	INT
Wishlist	WishlistID	INT
AuditLog	LogID	INT

3. Foreign Key References

Entity	Foreign Key	References
Address	CustomerID	Customer
Product	CategoryID	Category
ProductImage	ProductID	Product
Cart	CustomerID, ProductID	Customer, Product
Order	CustomerID, ShippingAddressID	Customer, Address
OrderItem	OrderID, ProductID	Order, Product
Payment	OrderID	Order
Shipment	OrderID	Order
Review	CustomerID, ProductID	Customer, Product
Wishlist	CustomerID, ProductID	Customer, Product
Category	ParentCategoryID	Category (self)

4. Key Business Rules

Customer: one customer can have multiple addresses, multiple orders, and multiple reviews.

Product: one product can have multiple images and multiple reviews, but belongs to exactly one category.

Order: one order can have multiple order items, but exactly one payment and one shipment.

Payment: an order is confirmed only after payment is successful; each order has exactly one payment record.

Inventory: stock quantity cannot be negative; a low-stock alert is raised when stock falls below the reorder level.

Audit: all admin actions are logged; customer accounts are marked inactive rather than deleted.